Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.22)

Metric

$$ds^2 = \left(-\frac{m^2 - 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\right) \mathrm{d}t \otimes \mathrm{d}t + \left(\frac{m^4 + 8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right) + 24\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 32\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3 + 16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}{16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}\right) \mathrm{d}\bar{x}$$
$$\otimes \mathrm{d}\bar{x} + \left(\frac{m^4 + 8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right) + 24\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 32\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3 + 16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}{16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}\right) \mathrm{d}\bar{y}$$
$$\otimes \mathrm{d}\bar{y} + \left(\frac{m^4 + 8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right) + 24\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 32\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3 + 16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}{16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}\right) \mathrm{d}\bar{z} \otimes \mathrm{d}\bar{z}$$

Coordinates

$(x^\mu) = (t, \bar{x}, \bar{y}, \bar{z})$

Null Tetrad

Covariant components

$$l = \left(\frac{2\sqrt{\frac{1}{2}}}{\frac{m}{h(\bar{x},\bar{y},\bar{z})} + 2} - \frac{\sqrt{\frac{1}{2}}m}{\left(\frac{m}{h(\bar{x},\bar{y},\bar{z})} + 2\right)h\left(\bar{x},\bar{y},\bar{z}\right)}\right)\, dt + \left(-\sqrt{\frac{1}{2}} - \frac{\sqrt{\frac{1}{2}}m^2}{4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2} - \frac{\sqrt{\frac{1}{2}}m}{h\left(\bar{x},\bar{y},\bar{z}\right)}\right)\, d\bar{x}$$

$$n = \left(\frac{2\sqrt{\frac{1}{2}}}{\frac{m}{h(\bar{x},\bar{y},\bar{z})} + 2} - \frac{\sqrt{\frac{1}{2}}m}{\left(\frac{m}{h(\bar{x},\bar{y},\bar{z})} + 2\right)h\left(\bar{x},\bar{y},\bar{z}\right)}\right)\, dt + \left(\sqrt{\frac{1}{2}} + \frac{\sqrt{\frac{1}{2}}m^2}{4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2} + \frac{\sqrt{\frac{1}{2}}m}{h\left(\bar{x},\bar{y},\bar{z}\right)}\right)\, d\bar{x}$$

$$m = \left(\sqrt{\frac{1}{2}} + \frac{\sqrt{\frac{1}{2}}m^2}{4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2} + \frac{\sqrt{\frac{1}{2}}m}{h\left(\bar{x},\bar{y},\bar{z}\right)}\right)\, d\bar{y} + \left(i\sqrt{\frac{1}{2}} + \frac{i\sqrt{\frac{1}{2}}m^2}{4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2} + \frac{i\sqrt{\frac{1}{2}}m}{h\left(\bar{x},\bar{y},\bar{z}\right)}\right)\, d\bar{z}$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}} + \frac{\sqrt{\frac{1}{2}}m^2}{4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2} + \frac{\sqrt{\frac{1}{2}}m}{h\left(\bar{x},\bar{y},\bar{z}\right)}\right) d\bar{y} + \left(-i\sqrt{\frac{1}{2}} - \frac{i\sqrt{\frac{1}{2}}m^2}{4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2} - \frac{i\sqrt{\frac{1}{2}}m}{h\left(\bar{x},\bar{y},\bar{z}\right)}\right) d\bar{z}$$

Contravariant components

$$l = \left(\frac{m}{\sqrt{2}m - 2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)} + \frac{2\,h\left(\bar{x},\bar{y},\bar{z}\right)}{\sqrt{2}m - 2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)}\right) dt + \left(-\frac{2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\right) d\bar{x}$$

$$n = \left(\frac{m}{\sqrt{2}m - 2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)} + \frac{2\,h\left(\bar{x},\bar{y},\bar{z}\right)}{\sqrt{2}m - 2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)}\right) dt + \left(\frac{2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\right) d\bar{x}$$

$$m = \left(\frac{2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\right) d\bar{y} + \left(\frac{2i\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\right) d\bar{z}$$

$$\overline{m} = \left(\frac{2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\right) d\bar{y} + \left(-\frac{2i\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\right) d\bar{z}$$

Directional Derivatives

$$D = \frac{m}{\sqrt{2}m - 2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)} + \frac{2\,h\left(\bar{x},\bar{y},\bar{z}\right)}{\sqrt{2}m - 2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)}\,\partial_t - \frac{2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\,\partial_{\bar{x}}$$

$$\Delta = \frac{m}{\sqrt{2}m - 2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)} + \frac{2\,h\left(\bar{x},\bar{y},\bar{z}\right)}{\sqrt{2}m - 2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)}\,\partial_t + \frac{2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\,\partial_{\bar{x}}$$

$$\delta = \frac{2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\,\partial_{\bar{y}} + \frac{2i\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\,\partial_{\bar{z}}$$

$$\overline{\delta} = \frac{2\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\,\partial_{\bar{y}} - \frac{2i\sqrt{2}h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^2 + 4\,mh\left(\bar{x},\bar{y},\bar{z}\right) + 4\,h\left(\bar{x},\bar{y},\bar{z}\right)^2}\,\partial_{\bar{z}}$$

Spin Coefficients

$$\rho = -\frac{4\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{x}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^3+6\,m^2h\left(\bar{x},\bar{y},\bar{z}\right)+12\,mh\left(\bar{x},\bar{y},\bar{z}\right)^2+8\,h\left(\bar{x},\bar{y},\bar{z}\right)^3}$$

$$\sigma = 0$$

$$\kappa = -\frac{2\sqrt{2}m^2h\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{y}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}+\frac{8\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)^2\frac{\partial}{\partial\bar{y}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}$$
$$-\frac{2i\sqrt{2}m^2h\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{z}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}+\frac{8i\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)^2\frac{\partial}{\partial\bar{z}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}$$

$$\mu = -\frac{4\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{x}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^3+6\,m^2h\left(\bar{x},\bar{y},\bar{z}\right)+12\,mh\left(\bar{x},\bar{y},\bar{z}\right)^2+8\,h\left(\bar{x},\bar{y},\bar{z}\right)^3}$$

$$\lambda = 0$$

$$\nu = \frac{2\sqrt{2}m^2h\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{y}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}-\frac{8\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)^2\frac{\partial}{\partial\bar{y}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}$$
$$-\frac{2i\sqrt{2}m^2h\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{z}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}+\frac{8i\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)^2\frac{\partial}{\partial\bar{z}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}$$

$$\alpha = \frac{2\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{y}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^3+6\,m^2h\left(\bar{x},\bar{y},\bar{z}\right)+12\,mh\left(\bar{x},\bar{y},\bar{z}\right)^2+8\,h\left(\bar{x},\bar{y},\bar{z}\right)^3}-\frac{2i\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{z}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^3+6\,m^2h\left(\bar{x},\bar{y},\bar{z}\right)+12\,mh\left(\bar{x},\bar{y},\bar{z}\right)^2+8\,h\left(\bar{x},\bar{y},\bar{z}\right)^3}$$

$$\beta = -\frac{2\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{y}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^3+6\,m^2h\left(\bar{x},\bar{y},\bar{z}\right)+12\,mh\left(\bar{x},\bar{y},\bar{z}\right)^2+8\,h\left(\bar{x},\bar{y},\bar{z}\right)^3}-\frac{2i\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{z}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^3+6\,m^2h\left(\bar{x},\bar{y},\bar{z}\right)+12\,mh\left(\bar{x},\bar{y},\bar{z}\right)^2+8\,h\left(\bar{x},\bar{y},\bar{z}\right)^3}$$

$$\tau = \frac{2\sqrt{2}m^2h\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{y}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}+\frac{2i\sqrt{2}m^2h\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{z}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}$$

$$\pi = -\frac{2\sqrt{2}m^2 h\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{y}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}+\frac{2i\sqrt{2}m^2 h\left(\bar{x},\bar{y},\bar{z}\right)\frac{\partial}{\partial\bar{z}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}$$

$$\epsilon = \frac{4\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)^2\frac{\partial}{\partial\bar{x}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}$$

$$\gamma = \frac{4\sqrt{2}mh\left(\bar{x},\bar{y},\bar{z}\right)^2\frac{\partial}{\partial\bar{x}}h\left(\bar{x},\bar{y},\bar{z}\right)}{m^4+4\,m^3h\left(\bar{x},\bar{y},\bar{z}\right)-16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3-16\,h\left(\bar{x},\bar{y},\bar{z}\right)^4}$$

Ricci Tensor Components

$$
\begin{aligned}
\Phi_{00} = &-\frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+ \frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+ \frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{x})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&- \frac{32\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{x})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&- \frac{24\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+ \frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+ \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&- \frac{32\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&- \frac{24\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+ \frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+ \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&- \frac{32\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}
\end{aligned}
$$

$$\Phi_{01} = -\frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial\bar{x}\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$
$$- \frac{8i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial\bar{x}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$
$$- \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$
$$- \frac{8i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$

$$\Phi_{02} = \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$
$$+ \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial\bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$
$$+ \frac{16i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial\bar{y}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$
$$+ \frac{16i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$
$$- \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$
$$- \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial\bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$

$$\Phi_{10} = -\frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial\bar{x}\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$
$$+ \frac{8i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial\bar{x}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$
$$- \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$
$$+ \frac{8i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}$$

$$
\begin{aligned}
\Phi_{11} = &-\frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{32\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&-\frac{16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{x})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&-\frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{32\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&-\frac{16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&-\frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{32\,mh\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&-\frac{16\,mh\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}
\end{aligned}
$$

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\begin{aligned}
\Phi_{12} = &\frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial \bar{x}\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{8i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial \bar{x}\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{8i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{20} = & \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
& + \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
& - \frac{16 i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial \bar{y} \partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
& - \frac{16 i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
& - \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
& - \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{21} = & \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial \bar{x} \partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
& - \frac{8 i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial \bar{x} \partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
& + \frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
& - \frac{8 i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6 + 8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right) + 20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2 - 80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{22} = &-\frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{x})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&-\frac{32\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{x})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&-\frac{24\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&-\frac{32\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&-\frac{24\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&+\frac{8\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6} \\
&-\frac{32\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6}
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = &-\frac{32\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)} \\
&+\frac{32\,m h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)} \\
&+\frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{x})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)} \\
&-\frac{16\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{x})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)} \\
&-\frac{32\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)} \\
&+\frac{32\,m h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)} \\
&+\frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)} \\
&-\frac{16\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)} \\
&-\frac{32\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)} \\
&+\frac{32\,m h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)} \\
&+\frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)} \\
&-\frac{16\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^6+8\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)+20\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^2-80\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4-128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5-64\,h\left(\bar{x},\bar{y},\bar{z}\right)^6\right)}
\end{aligned}
$$

Weyl Tensor Components

$$
\begin{aligned}
\Psi_0 = & \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{112\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial\bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial\bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{(\partial\bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{16i\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial\bar{y}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{32i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{\partial\bar{y}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{128i\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{\partial\bar{y}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{16i\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{224i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{256i\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7}
\end{aligned}
$$

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\begin{aligned}
&- \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&+ \frac{112\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&- \frac{128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&- \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&+ \frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&+ \frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7}
\end{aligned}
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\begin{aligned}
\Psi_1 = & \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial\bar{x}\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{\partial\bar{x}\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{64\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{\partial\bar{x}\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{8i\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial\bar{x}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{16i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{\partial\bar{x}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{64i\,mh\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{\partial\bar{x}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{112\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{128\,mh\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{8i\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{112i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{128i\,mh\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,mh\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7}
\end{aligned}
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\begin{aligned}
\Psi_2 = & \frac{16\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
& - \frac{224\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
& + \frac{256\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial \bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
& + \frac{16\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{x})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
& - \frac{32\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{x})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
& - \frac{128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{(\partial \bar{x})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
& - \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
& + \frac{112\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
& - \frac{128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
& - \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
& + \frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
& + \frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)}
\end{aligned}
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\begin{aligned}
&- \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
&+ \frac{112\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
&- \frac{128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
&- \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
&+ \frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)} \\
&+ \frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{3\left(m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Psi_3 = &-\frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial\bar{x}\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&+\frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{\partial\bar{x}\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&+\frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{\partial\bar{x}\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&+\frac{8i\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial\bar{x}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&-\frac{16i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{\partial\bar{x}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&-\frac{64i\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{\partial\bar{x}\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&-\frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&+\frac{112\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&-\frac{128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&+\frac{8i\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&-\frac{112i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
&+\frac{128i\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial\bar{x}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial\bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7+10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right)+36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2+40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3-80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4-288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5-320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6-128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7}
\end{aligned}
$$

$$
\begin{aligned}
\Psi_4 = & \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{112\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{(\partial \bar{y})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{16i\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{\partial \bar{y}\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{32i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{\partial \bar{y}\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{128i\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{\partial \bar{y}\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{16i\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& + \frac{224i\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7} \\
& - \frac{256i\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial \bar{y}} h\left(\bar{x},\bar{y},\bar{z}\right) \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7}
\end{aligned}
$$

$$- \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^2 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7}$$

$$+ \frac{112\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7}$$

$$- \frac{128\,m h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial}{\partial \bar{z}} h\left(\bar{x},\bar{y},\bar{z}\right)^2}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7}$$

$$- \frac{8\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^3 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7}$$

$$+ \frac{16\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^4 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7}$$

$$+ \frac{64\,m h\left(\bar{x},\bar{y},\bar{z}\right)^5 \frac{\partial^2}{(\partial \bar{z})^2} h\left(\bar{x},\bar{y},\bar{z}\right)}{m^7 + 10\,m^6 h\left(\bar{x},\bar{y},\bar{z}\right) + 36\,m^5 h\left(\bar{x},\bar{y},\bar{z}\right)^2 + 40\,m^4 h\left(\bar{x},\bar{y},\bar{z}\right)^3 - 80\,m^3 h\left(\bar{x},\bar{y},\bar{z}\right)^4 - 288\,m^2 h\left(\bar{x},\bar{y},\bar{z}\right)^5 - 320\,m h\left(\bar{x},\bar{y},\bar{z}\right)^6 - 128\,h\left(\bar{x},\bar{y},\bar{z}\right)^7}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"